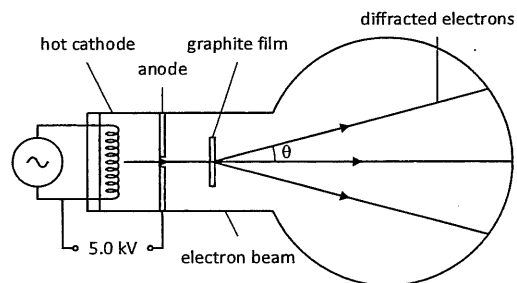


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/26]

A diagram shows a simplified setup of electron tube to demonstrate electron diffraction. The electrons are diffracted by angle θ . Diffraction rings appear on the screen.



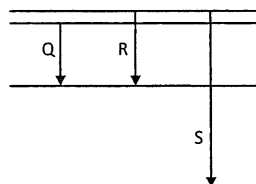
The potential difference between the cathode and anode is increased from 5.0 kV to 10 kV.

Which of the following statement is false?

- A The velocity of electrons hitting the graphite film increases.
- B The diffracted angle θ decreases.
- C The brightness of the central ring increases.
- D The de Broglie wavelength of the electrons from the anode increases.

2. [AJC/H2/Prelims 2018/P1/27]

The diagram below shows some of the energy levels of an atom. The three transitions shown result in radiation of ultraviolet (UV), as well as green and yellow regions of the visible light spectrum.

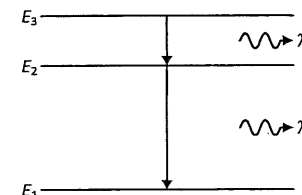


Which one of the choices below correctly identifies the transitions?

	Ultraviolet	Green visible light	Yellow visible light
A	S	R	Q
B	S	Q	R
C	Q	R	S
D	Q	S	R

3. [CJC/H2/Prelims 2018/P1/28]

The diagram below shows a simplified representation of the three electron energy levels in an atom.



Cool vapour of this element at low pressure is bombarded with electrons accelerated from rest across a potential difference V . Two possible transitions which result in the emission of photons of wavelengths $\lambda_1 = 6.22 \times 10^{-7} \text{ m}$ and $\lambda_2 = 1.78 \times 10^{-7} \text{ m}$ are observed.

What is the minimum value of V for the above transitions to occur?

- A 1.56 V
- B 2.80 V
- C 7.00 V
- D 9.00 V

4. [DHS/H2/Prelims 2018/P1/28]

A proton has a kinetic energy of 1.00 MeV.

If its momentum is measured with an uncertainty of 1.00%, what is the minimum uncertainty in its position?

- A $5.64 \times 10^{-14} \text{ m}$
- B $9.08 \times 10^{-14} \text{ m}$
- C $2.87 \times 10^{-12} \text{ m}$
- D $9.77 \times 10^{-10} \text{ m}$

5. [IJC/H2/Prelims 2018/P1/27]

A photon is emitted from a hydrogen atom by an electron transition between two energy levels. The energy levels have energies X and Y .

Which expression gives the momentum of this photon?

- A $\frac{(X-Y)}{h}$
- B $\frac{(X-Y)}{c}$
- C $\frac{(X-Y)}{hc}$
- D $\frac{h(X-Y)}{c}$

6. [IJC/H2/Prelims 2018/P1/28]

A proton has a kinetic energy of 1.00 MeV.

If its momentum is measured with an uncertainty of 1.00%, what is the minimum uncertainty in its position?

- A $1.22 \times 10^{-10} \text{ m}$
- B $2.87 \times 10^{-12} \text{ m}$
- C $9.08 \times 10^{-12} \text{ m}$
- D $6.59 \times 10^{-15} \text{ m}$

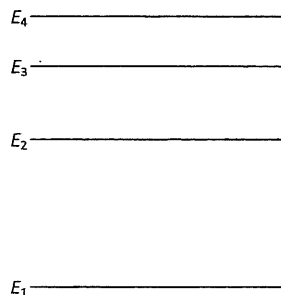
7. [JJC/H2/Prelims 2018/P1/27]

Which type of electromagnetic radiation is emitted when an electron in an atom makes a transition from an energy level at -1.5 eV to an energy level at -3.5 eV ?

- A Microwaves
B Infra-red
C Visible light
D Ultraviolet

8. [MJC/H2/Prelims 2018/P1/26]

The diagram represents in simplified form some of the energy levels of the hydrogen atom.



The transition of an electron from E_3 to E_2 is associated with the emission of red light.

Which transition could be associated with the emission of blue light?

- A E_4 to E_1 B E_1 to E_4 C E_4 to E_2 D E_2 to E_4

9. [MJC/H2/Prelims 2018/P1/27]

An electron has a kinetic energy of 1.0 MeV .

If its momentum is measured with an uncertainty of 1.0% , what is the uncertainty in its position?

- A $7.7 \times 10^{-30} \text{ m}$ B $1.2 \times 10^{-10} \text{ m}$ C $2.9 \times 10^{-12} \text{ m}$ D $4.1 \times 10^{-19} \text{ m}$

10. [NJC/H2/Prelims 2018/P1/27]

In the lungs, there are tiny sacs of air known as alveoli. The average diameter of an alveolus is 0.250 mm . Consider an oxygen molecule of mass $5.30 \times 10^{-26} \text{ kg}$ that is trapped in an alveolus.

What is the order of magnitude of the uncertainty in the velocity of this oxygen molecule?

- A 10^{-5} m s^{-1} B 10^{-8} m s^{-1} C $10^{-10} \text{ m s}^{-1}$ D $10^{-12} \text{ m s}^{-1}$

11. [NYJC/H2/Prelims 2018/P1/27]

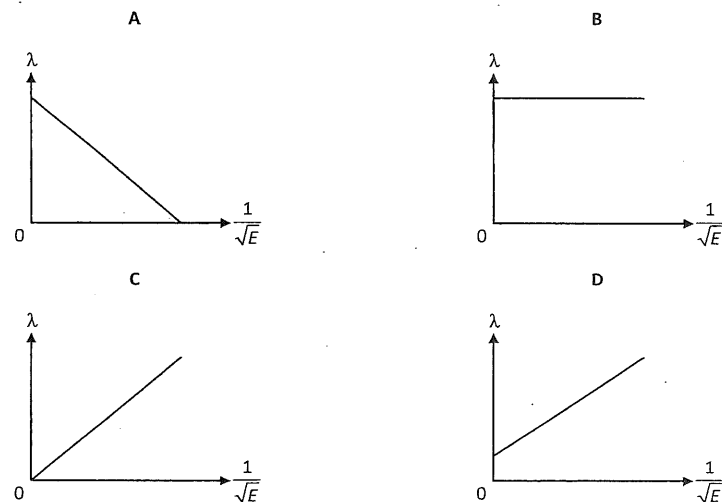
What is the de Broglie wavelength of an alpha particle of kinetic energy E , given that u is the unified atomic mass unit?

- A $\frac{h}{2\sqrt{uE}}$ B $\frac{2\sqrt{uE}}{h}$ C $\frac{h}{2\sqrt{2uE}}$ D $\frac{2\sqrt{2uE}}{h}$

12. [PJC/H2/Prelims 2018/P1/27 Modified]

An electron with kinetic energy E has a de Broglie wavelength of λ .

Which of the following graphs correctly represents the relationship between λ and E ?



13. [PIC/H2/Prelims 2018/P1/28]

White light from a filament lamp is passed through cooled hydrogen gas and viewed through a prism.

Which of the following best describes the spectrum?

- A Dark lines on a coloured background
B Dark lines on a white background
C Coloured lines on a black background
D Coloured lines on a white background

14. [RVHS/H2/Prelims 2018/P1/27]

The velocity of a proton is twice the velocity of an electron.

Assuming the mass of a proton is approximately 2000 times that of the mass of an electron, what is the ratio of the de Broglie wavelength of the proton to that of the electron?

- A 2.5×10^{-4} B 0.50 C 2.0 D 4000

15. [SRJC/H2/Prelims 2018/P1/28]

The momentum of an alpha particle is measured with an uncertainty of 2.0% .

Given that it has a kinetic energy of 1.00 MeV , what is the minimum uncertainty in its position?

- A $2.1 \times 10^{-19} \text{ m}$ B $7.2 \times 10^{-13} \text{ m}$ C $1.6 \times 10^{-12} \text{ m}$ D $7.2 \times 10^{-10} \text{ m}$

16. [YJC/H2/Prelims 2018/P1/29]

The transition of electrons between three consecutive energy levels in a particular atom gives rise to three spectral lines. The shortest and longest wavelengths of those spectral lines are λ_1 and λ_2 respectively.

The wavelength of the other spectral line is

- A $\frac{\lambda_1 + \lambda_2}{2}$ B $\lambda_1 - \lambda_2$ C $\frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$ D $\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)^{-1}$

II. Structured Questions (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P3/5]

- a. Fig. 1.1 shows a gold-leaf electroscope. The gold leaf is deflected because the electroscope is negatively charged.

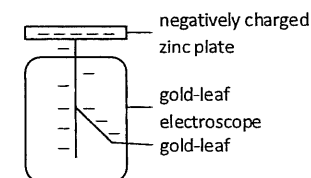


Fig. 1.1

- i. When red light is incident on the electroscope, the gold leaf remains deflected as shown in Fig. 1.2(a). When ultra-violet radiation is incident on the electroscope, the gold leaf falls as shown in Fig. 1.2(b). This is due to a decrease in the negative charge of the gold leaf.

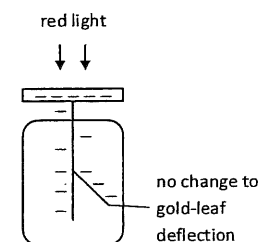


Fig. 1.2(a)

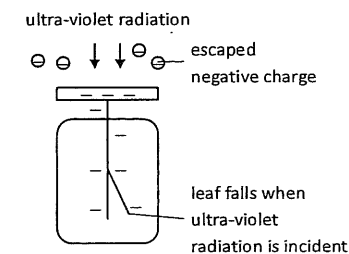


Fig. 1.2(b)

Using Fig. 1.2(a) and Fig. 1.2(b), explain how this is a demonstration of the photoelectric effect. [2]

- ii. When the intensity of the red light that is incident on the electroscope in Fig. 1.2(a) is increased, the deflection of the gold leaf remains the same.

Explain how this is evidence for the particulate nature of electromagnetic radiation. [2]

- b. An electron travelling at 1200 m s^{-1} passes through a slit of $24 \mu\text{m}$ as shown in Fig. 1.3.

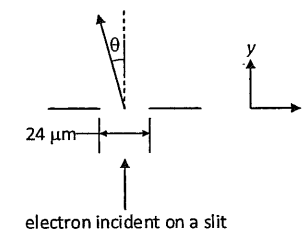


Fig. 1.3

- i. Calculate the minimum uncertainty of the momentum of the electron in the x-direction, Δp_x . [2]

- ii. In an experiment where many electrons are put through the slit, it is observed that most electrons deviate from its original path within the angle θ as shown in Fig. 1.3 due to the uncertainty of the electrons' momentum.

Use your answer in b.i. or otherwise, calculate θ .

[3]

[Total: 9]

2. [AJC/H2/Prelims 2018/P3/8 Modified]

- c. Einstein's observation introduced the particulate nature of light waves.

- Describe an experiment that shows the wave nature of a particle. [2]
- Calculate the de Broglie wavelength for a single electron which has a speed of $2.5 \times 10^6 \text{ m s}^{-1}$. [1]
- State and explain whether electrons with speed of $2.5 \times 10^6 \text{ m s}^{-1}$ are suitable to be used to study atomic structure of matter. [1]
- Neutrons can also be used to investigate the atomic structure of matter. Suggest and explain an advantage and a disadvantage in using neutrons. [3]

[Total: 7]

3. [CJC/H2/Prelims 2018/P2/8]

- State what is meant by the *photoelectric effect*. [1]
 - Describe the principal features that are observed in the photoelectric effect that support the particulate nature of electromagnetic radiation. [3]
- A low pressure sodium lamp produces an intensity of 0.20 W m^{-2} of yellow light of frequency $4.55 \times 10^{14} \text{ Hz}$ at a distance of 5.0 m from the lamp.
 - Assuming that the lamp acts a point source, show that the intensity a distance 20 m from the lamp is 0.013 W m^{-2} . [1]
 - Estimate the number of photons per second that would strike a piece of A4 size paper which is held 20 m perpendicular to the ray of light from the lamp. [3]

[Total: 8]

4. [HCI/H2/Prelims 2018/P3/5]

- a. Ernest Rutherford proposed a planetary model for the hydrogen atom. In the model, a single electron is treated as a point-like charged particle, moving in circular motion around a stationary proton (the nucleus) as shown in Fig. 4.1.

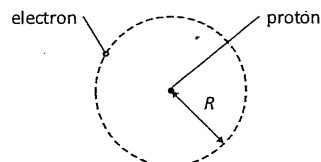


Fig. 4.1 (not to scale)

- State two forms of energy that the system possesses and state whether each form is by convention, positive or negative. [2]
- Explain why the total energy of the system is conventionally taken to be negative when the electron is orbiting the nucleus. [2]

- b. In quantum mechanics, it is not possible to model the path of the electron in the hydrogen atom to be a well-defined circle around the proton.
- Using Heisenberg uncertainty principle for position and momentum, explain why the path of the electron inside the atom is not well-defined. [2]
 - Despite not knowing the path of the electron, we are certain that the electron remains bound to the nucleus. By considering your answer to a., show that the maximum kinetic energy of the bound electron is given by

$$E_k = \frac{e^2}{4\pi\epsilon_0 R}$$

Explain your working clearly.

[2]

[Total: 8]

5. [SRJC/H2/Prelims 2018/P2/5]

- For a particular gas, the emission and absorption spectra are obtained for the visible light spectrum. Discuss one similarity and one difference between the discrete lines of the absorption and emission spectra of this gas. [2]
- Fig. 5.1 gives information on three lines observed in the emission spectrum of hydrogen atoms.

wavelength / nm	energy of photon / eV
486	2.56
656	
1880	0.66

Fig. 5.1

- Complete Fig. 5.1 by calculating the energy of the photon with wavelength 656 nm. [1]
- Fig. 5.2 is a partially completed diagram to show energy levels of a hydrogen atom.

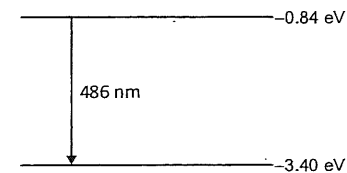


Fig. 5.2

On Fig. 5.2, draw an additional labelled energy level which will account for the emission of the photons in Fig. 5.1. [3]

- c. Three of the energy levels of a lithium atom are shown in Fig. 5.3.

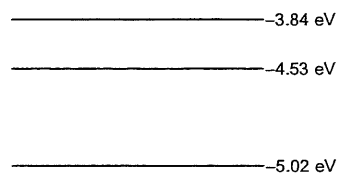


Fig. 5.3

One way to study the energy levels of an atom is to bombard the atom with electrons and measure the kinetic energies of the bombarding electrons before and after the collision. If a lithium atom which is originally in the -5.02 eV level is bombarded with an electron of kinetic energy 0.92 eV, the scattered electron can have only two possible kinetic energies.

States these two kinetic energy values, and state what happens to the lithium atom in each case.

[4]

[Total: 10]

6. [TPJC/H2/Prelims 2018/P2/6]

- a. Fig. 6.1 shows some energy levels for the hydrogen atom.

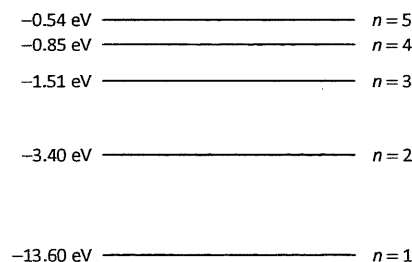


Fig. 6.1 (not to scale)

- State the number of emission lines that could be produced from the energy levels shown in Fig 6.1. [1]
 - Visible light wavelengths range approximately from 400 nm to 750 nm. Show that the emission spectrum when electrons make transitions to the $n=1$ state does not fall within the visible region of the electromagnetic spectrum. [3]
 - Hence or otherwise, determine the number of *visible* emission lines for the energy levels shown in Fig. 14.1. Show all workings clearly. [4]
- b. The atomic energy levels for a multi-electron atom can be approximated using the formula

$$E_n = -\frac{13.60}{n^2} (Z-1)^2$$

where Z is the atomic number of the atom, n is the quantum number of the shell, and E_n is the energy of the shell in eV.

- Using the formula provided in b., determine the energy of the photon released when an electron transitions from $n=2$ to $n=1$ in a tungsten atom (atomic number 74). [3]
- Hence state the type of electromagnetic radiation emitted when an electron transitions from $n=2$ to $n=1$ in a tungsten atom. [1]

[Total: 12]

7. [TPJC/H2/Prelims 2018/P3/6]

- State what is meant by a *photon*. [1]
- An X-ray photon of wavelength 965.0 pm collides elastically with a stationary electron, as shown in Fig 7.1.

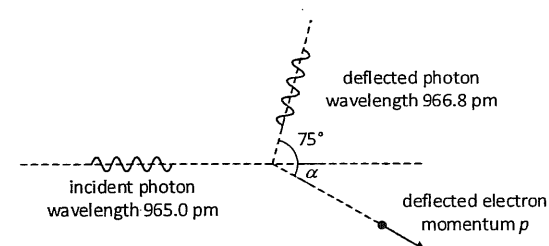


Fig. 7.1

The photon is deflected through an angle of 75° and its wavelength changes to 966.8 pm after the collision. The electron is deflected through an angle α .

- Calculate the change in the energy of the photon as it is deflected. [2]
 - Use conservation of energy to show that the momentum p of the deflected electron is 8.36×10^{-25} N s. [2]
- c. Momentum is a vector quantity. By taking components of momentum perpendicular to the direction of the incident photon, calculate the angle α of deflection of the electron as shown in Fig 7.1. [5]

Explain your working.

[Total: 10]

8. [VJC/H2/Prelims 2018/P3/7]

- In an electron microscope, electrons are accelerated by an electric field in order to investigate the structure of matter. This is done by analyzing the diffraction pattern produced when the accelerated electron beam is passed through the arrangement of atoms in solids. This technique is possible due to the phenomenon known as wave-particle duality.
 - Explain what is meant by the term *wave-particle duality* as applied to this technique. [1]
 - Estimate the kinetic energy of electrons such that they can be used to measure the distance between adjacent atoms in crystals, which is typically about 0.1 nm. [4]
 - The percentage uncertainty in measuring the electron's velocity is 10% . Calculate the minimum uncertainty in a simultaneous measurement of the electron's position. [2]
 - Calculate the energy of a photon which has the same wavelength as the electron. [2]
- Scientists suggest the use of protons to replace the electrons in the electron microscope in part a.
 - Determine the associated wavelength of protons with the same kinetic energy as the electrons in part a. [1]
 - State and explain whether the use of protons would make a better or less effective probe of small-scale structures. [1]

[Total: 11]

9. [YJC/H2/Prelims 2018/P2/6 Modified]

- c. Fig. 9.3 shows the experimental set-up for an electron diffraction experiment.

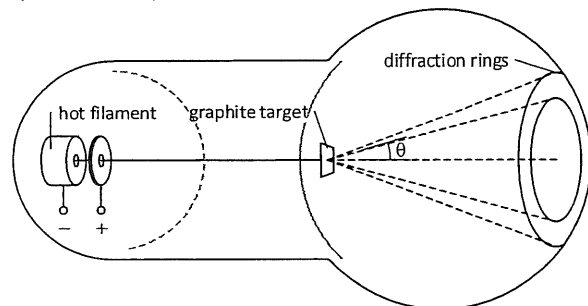


Fig. 9.3

Bright rings appear on the curved screen. It was found that the first off-centre maximum (ring) occurred at $\theta = 9.96^\circ$ for the accelerating potential of 625 V.

The diffraction of the electron beam is in accordance with the following expression

$$2d \sin \theta = n\lambda_{\text{dB}}$$

whereby d is the inter-atomic distance in the target, n the order number and λ_{dB} the de Broglie wavelength of an electron in the beam.

- i. Determine the values of

1. λ_{dB} , and

[2]

2. d .

[2]

- ii. State and explain what happens to the rings when the accelerating potential is decreased.

[4]

[Total: 8]